

Synthesis of 5-Pyridyl-2-furaldehydes via Palladium-Catalyzed Cross-Coupling with Triorganozincates

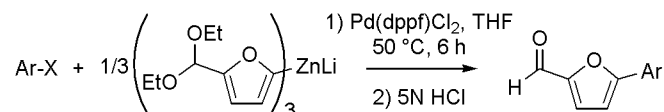
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ABSTRACT



5-Pyridyl- and 5-aryl-2-furaldehydes are prepared from furaldehyde diethyl acetal in a four-step, one-pot procedure: (i) deprotonation; (2) Li to Zn transmetalation; (3) Pd-mediated cross-coupling; (4) aldehyde deprotection. Triorganozincate 7 was found to transfer all three groups in the Pd-catalyzed cross-coupling reaction with haloaromatics.

Substituted furans are ubiquitous structural units in natural products and in pharmaceuticals¹ and have often been used as synthetic intermediates.² We are interested in 5-pyridyl-2-furaldehydes as synthetic intermediates for pharmaceutically active compounds. Herein we report a scaleable, operationally simple procedure to prepare 5-aryl- and 5-pyridyl-2-furaldehydes from inexpensive, commercially available 2-furaldehyde diethyl acetal.³ We also report that triorgano- and tetraorganozincates transfers all organic groups in the Pd-catalyzed cross-coupling reaction with aryl halides.

Several transition metal catalyzed biaryl coupling approaches to 5-pyridyl-2-furaldehydes were considered. The ease in which substitution can be incorporated at the 2- and 5-positions made this approach more attractive than cyclization of acyclic precursors.⁴ Although 2-halofurans are suitable biaryl coupling partners,⁵ the prohibitive cost of the requisite

5-bromo-2-furaldehyde⁶ precluded its use on large scale. Thus, we focused instead on using furaldehyde as the nucleophilic component. In a recent report, this strategy was employed using a Heck coupling to prepare 5-aryl-2-furfurals; unfortunately, 2-halopyridines were poor substrates.⁷ O'Doherty recently focused on Stille and Negishi couplings to prepare 5-aryl-2-furfurals.⁵ A similar approach to 5-pyridyl-2-furaldehydes using a one-pot, four-step sequence is illustrated in Scheme 1: (i) selective deprotonation of furaldehyde diethyl acetal (**1**); (ii) transmetalation to an organometallic nucleophile capable of undergoing transition metal catalyzed cross-coupling;⁸ (iii) cross-coupling; (iv) aldehyde deprotection.

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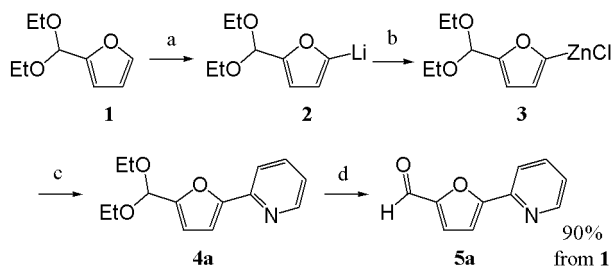
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(3) Available from Aldrich Chemical Co. at \$147/25 g, although the cost of raw material 2-furaldehyde is comparable to common solvents.

Scheme 1. Four-Step, One-Pot Synthesis of 5-Pyridylfuraldehydes^a



^a Reagents and conditions: (a) *n*-BuLi, THF, TMEDA, $-25\text{ }^{\circ}\text{C}$; (b) ZnCl_2 , $-25\text{ }^{\circ}\text{C}$ to rt; (c) 2-chloropyridine (**9a**), $\text{Pd}(\text{dppf})\text{Cl}_2$, 6 h, $50\text{ }^{\circ}\text{C}$; (d) 5 N HCl.

The deprotonation of furaldehyde diethyl acetal **1** in THF with *n*-BuLi is selective at $-78\text{ }^{\circ}\text{C}$, providing furyllithium **2** (Scheme 1).^{9,10} The extreme cryogenics of the deprotonation could be avoided with the addition of *N,N,N',N'*-tetramethylethylenediamine (TMEDA).¹¹ Using TMEDA, the deprotonation and subsequent transmetalation steps could be conducted at -20 to $-30\text{ }^{\circ}\text{C}$.^{12,13}

Organozinc reagents have seen widespread use due to exceptional functional group tolerance, ease of preparation, and removal of zinc byproducts.¹⁴ Treatment of **2** with ZnCl_2 (1.1 equiv) provided organozinc chloride **3**. Several catalysts derived from Ni and Pd were screened to optimize the reaction of **3** with 2-chloropyridine (**9a**). $\text{Pd}(\text{dppf})\text{Cl}_2$ ¹⁵ (2 mol %) gave the best results, providing **5a** in 90% yield after deprotection. For kilogram-scale runs, catalyst loading was reduced to 0.25 mol % without adverse effects on yield or reaction time.

Commercial bulk sources of “anhydrous” ZnCl_2 solid typically contain no less than 1% H_2O .¹⁶ Dissolution in THF and extensive molecular sieve drying was therefore necessary.¹⁷ For this reason, as well as reducing the overall reaction

volume and the amount of transition metal reagents,¹⁸ we aimed to minimize the amount of ZnCl_2 . Thus, diorganozinc **6** was initially prepared by simply adjusting the ratio of reagents (Figure 1). Using the appropriate furyllithium **2**/ ZnCl_2 stoichiometry, organozincate **7** (3:1) and higher order zincate **8** (4:1) were also prepared.

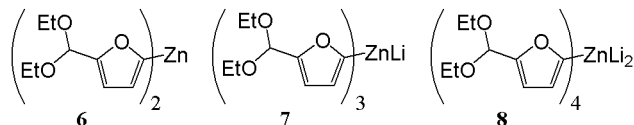


Figure 1.

Diorganozincs are more reactive than organozinc halides toward electrophiles and sometimes transfer only one of the two groups.¹⁹ In cross-coupling reactions, diorganozincs generally transfer both organic groups.²⁰ Mixed aryl–dialkyl zincates have been shown to selectively transfer an aryl group to an electrophile.^{21,22} We report here that triorganozincate **7** transfers all three organic groups in the cross-coupling reaction with 2-chloropyridine (**9a**) and is equally efficient as mono and bis organozincs **3** and **6** (Table 1, entries 1 and 4). However, slightly lower yields were typically obtained with higher order zincate **8** (Table 1, entry 5). We ascribe the lower yields with higher order zincate **8** to its lower thermal stability compared to that of **6** and **7**.²³

The optimized coupling procedure was applied using organozinc reagents **6** and **7** with chloropyridines **9a–g** (Table 1). Since we mainly were interested in chloropyridines, the coupling reaction was also conducted using the

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(10) The deprotonation step was monitored by HPLC by the following method: an aliquot of solution **2** was quenched with MeI. HPLC typically showed 95% conversion to **2** compared to authentic standards of furaldehyde and 5-methyl-2-furaldehyde.

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(12) Furyllithium **2** was stable in THF with TMEDA at $-20\text{ }^{\circ}\text{C}$ for several hours. Aging at higher temperatures (e.g., $-10\text{ }^{\circ}\text{C}$, 1 h) resulted in diminished yields of coupled product **5a** (54%).

(13) Without TMEDA, unidentifiable decomposition occurs upon aging the solution of **2** above $-40\text{ }^{\circ}\text{C}$.

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(16) Kilogram quantities of solid ZnCl_2 can be oven dried at $120\text{ }^{\circ}\text{C}$, 23 in. Hg with an N_2 purge for 3–5 days. However, handling of a THF solution on scale is preferred to the hygroscopic solid.

(17) Commercial ZnCl_2 solution (0.5 M in THF, Aldrich Chemical Co.) contains 0.5% H_2O ($\sim 0.3\text{ M}$) as determined by Karl Fisher titration.

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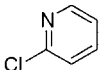
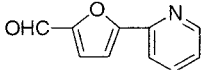
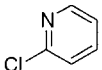
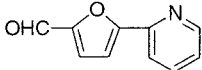
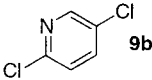
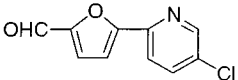
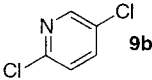
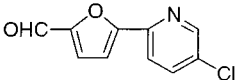
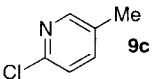
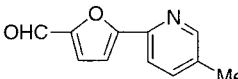
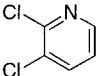
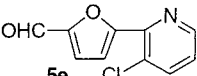
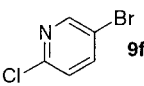
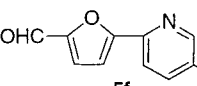
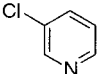
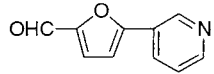
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(23) Yield of coupled product **5a** was significantly lower after aging **8** at $25\text{ }^{\circ}\text{C}$ for 1 h.

Table 1. Synthesis of 5-Substituted Pyridylfurals via Pd-Catalyzed Coupling of Pyridylhalide **9a-g** and Organozinc Reagents **6**, **7** and **8**

entry	substrate	Zn reagent ^a	catalyst	% yield	product
1	 9a	6	Pd(dppf)Cl ₂	92	 5a
2	 9a	6	1:1 Pd/(<i>t</i> -Bu) ₃ P ^b	91	 5a
3		7	1:1 Pd/(<i>t</i> -Bu) ₃ P ^b	<5	
4		7	Pd(dppf)Cl ₂	89	
5		8	Pd(dppf)Cl ₂	85	
6	 9b	6	Pd(dppf)Cl ₂	82	 5b
7	 9b	6	1:1 Pd/(<i>t</i> -Bu) ₃ P ^b	72 (8:1 5b/11)	 5b
8		7	Pd(dppf)Cl ₂	85	
9		6	Pd(dppf)Cl ₂	87	
10		6	Pd(dppf)Cl ₂	90	
11	 9c	7	Pd(dppf)Cl ₂	80	 5c
12		6	Pd(dppf)Cl ₂	81	
13		6	1:1 Pd/(<i>t</i> -Bu) ₃ P ^b	73	
14		6	1:1 Pd/(<i>t</i> -Bu) ₃ P ^b	58 (4:1)	
15	 9e	6	Pd(dppf)Cl ₂	36 (5e only)	 5e
16	 9f	6	1:1 Pd/(<i>t</i> -Bu) ₃ P ^b	70 (1:1)	 5f
17		6	Pd(dppf)Cl ₂	70 (1:1)	
18		6	1:1 Pd/(<i>t</i> -Bu) ₃ P ^b	79 (11 only)	
19		6	Pd(dppf)Cl ₂	30	
20	 9g	6	1:1 Pd/(<i>t</i> -Bu) ₃ P ^b	76	 5g
21		7	1:1 Pd/(<i>t</i> -Bu) ₃ P ^b	<5	
22		7	Pd(dppf)Cl ₂	<5	
23		7	Pd(dppf)Cl ₂	<5	

^a Organozinc reagents were used in the following stoichiometry: **6** (0.6 equiv), **7** (0.35 equiv), and **8** (0.27 equiv). ^b "1:1 Pd/(*t*-Bu)₃P" was prepared from a 0.5:1 ratio of Pd₂(dba)₃ to (*t*-Bu)₃P.

catalyst system derived from (*t*-Bu)₃P/Pd₂(dba)₃.²⁴ As discovered by Fu, the (*t*-Bu)₃P: Pd ratio is an important parameter.²⁵ For diorganozinc **6**, a 1:1 ratio gave significantly better yields than commercially available Pd(P(*t*-Bu)₃)₂.²⁶ Use of Pd₂(dba)₃ with monodentate phosphine ligands such as P(*t*-Bu)₃, P(*c*-Hx)₃, PPh₃, P(2-furyl)₃, and P(*o*-tolyl)₃ failed to promote the reaction of halopyridines or haloaromatics

with zincates **7** or **8** (Table 1, entries 3 and 21). Bidentate ligands, such as dppf, dppe, dppp, and dppb, promoted the Pd-catalyzed reaction with zincates **7** and **8**, but dppf was superior.

We also examined a series of aryl bromides in the Pd(dppf)Cl₂-catalyzed cross-coupling with organozinc re-

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(26) Coupling of **6** and **9g** with (*t*-Bu)₃P/Pd₂(dba)₃ (2 mol % Pd) at 0.5:1, 1:1, 2:1, and 3:1 (ligand: Pd) gave 70%, 76%, 20%, and 20% yields, respectively.

Table 2. Pd(dppf)Cl₂-Catalyzed Preparation of 5-Substituted Arylfurfurals from Aryl Halides **12a–f** and Organozinc Reagents **6**, **7**, and **8**

Reaction scheme: $\text{EtO-CH}_2\text{-furan-ZnLi}_x \xrightarrow[2) \text{ 5N HCl}]{1) \text{ 12a-f}} \text{H-C(=O)-furan-CH}_2\text{-Ar}$

6, 7, 8 ($X=0,1,2$) 13a-f

entry	substrate (R)	organozinc reagent ^a	yield (%) ^b
1	12a (H)	6	77
2		7	76
3		8	71
4	12b (Me)	6	88
5		7	78
6		8	76
7	12c (OMe)	6	85
8		7	79
9		8	75
10	12d (NO ₂)	6	75
11		7	62
12		8	41
13	12e (CN)	6	84
14		7	81
15		8	61
16	12f (CO ₂ Et)	6	84
17		7	63
18		8	56

^a Organozinc reagents were used in the following stoichiometry: **6** (0.6 equiv), **7** (0.35 equiv), and **8** (0.27 equiv). ^b HPLC assay yields base on an authentic standard of each product.

agents **6**, **7**, and **8** (Table 2). Making optimal use of the transition metal reagent ZnCl₂ via complex **8** is more efficient than standard Negishi couplings; however, it suffers from less functional group tolerance (Table 2, entries 12, 15, and 18).

Spectroscopic evidence for zincate and higher order zincate formation was relayed to a simpler system generated from PhLi. ¹³C NMR data indicates a clear trend (Table 3) in that a low-field shift in the ipso Ph-M carbon occurs with increasing electron density in the aromatic ring.²⁷ In the cross-coupling reaction with 2-chloropyridine, Ph₃ZnLi (0.35 equiv) and Ph₄ZnLi₂ (0.27 equiv) transferred all aryl groups

(27) This trend is opposite for aliphatic protons, see: Uchiyama, M.; Koike, M.; Kameda, M.; Kondo, Y.; Sakamoto, T. *J. Am. Chem. Soc.* **1996**, *118*, 8733–8734.

Table 3. Synthesis of 2-Phenylpyridine (**13a**) via Pd(dppf)Cl₂-Catalyzed Coupling of Ph-M with 2-Chloropyridine (1.2 equiv)

entry	δ _{Ph} (ppm) ^a	Ph-M (equiv)	yield (%)
1	187.4	PhLi (1.0) ^b	<5%
2	159.4	PhZnCl (1.0)	98
3	161.8	Ph ₂ Zn (0.6)	95
4	168.9	Ph ₃ ZnLi (0.35)	100
5	170.5	Ph ₄ ZnLi ₂ (0.27)	99

^a Ipso Ph carbon (M–C) ¹³C data in THF-*d*₈, with reference to C-2 THF signal at 67.57 ppm. ^b PhLi was prepared from bromobenzene and *s*-BuLi.

in nearly quantitative yield. To address whether the reaction is catalytic in Zn(II), ZnCl₂ (10 mol %) was added to PhLi, followed by 2-chloropyridine and Pd(dppf)Cl₂ (2 mol %). Only a 37% yield was attained, indicating the initial formation of Ph₄ZnLi₂ followed by transfer of all four groups.²⁸ Interestingly, slow addition of PhLi to a THF solution of ZnCl₂ (10 mol %), 2-chloropyridine, and Pd(dppf)Cl₂ (2 mol %) produced an 82% yield of **13a**.²⁹

An extremely efficient process for the synthesis of 5-aryl- and 5-pyridyl-2-furaldehydes has been demonstrated. The four-step, one-pot procedure takes advantage of using sub-stoichiometric quantities of ZnCl₂. A triorganozincate intermediate successfully transfers all three groups in the cross-coupling reaction. Thermally stable higher order zincates (e.g., Ph₄ZnLi₂) transfer all four groups. Efforts to understand and find further application of this chemistry are ongoing.

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Supporting Information Available: Experimental procedures and characterizations for all new compounds. This information is available free of charge via the Internet at <http://pubs.acs.org>.

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(28) The reaction appears to stall, and added Pd catalyst failed to produce more product.

(29) The PhLi (1 equiv) was added over 1 h at room temperature to 35 °C. The reaction was then aged at 60 °C for 2 h. Biphenyl (9%) was the only byproduct.